



Smishing Detection: Combating SMS Phishing Attacks by Utilizing Machine-Learning Algorithms

Aqsa Shaikh, Mariya Shaikh, Srivaramangai R.

Abstract: With the rapid uptake of mobile communications, cybercriminals have increasingly resorted to using SMS (Short Message Services) in the guise of phishing attacks commonly referred to as smishing (SMS phishing). Phishing SMS messages impersonate trusted organizations to persuade users into clicking malicious links, providing personal credentials, or installing malware. This paper reviews up-to-date advancements in machine learning for smishing detection, using insights derived from various studies on the subject. It looks into critical machine learning models such as Deep Learning models (CNN, LSTM), Logistic Regression, Random Forest, Support Vector Machines (SVM), and Gradient Boosting, to classify messages as spam, phishing, or legitimate. It examines feature extraction techniques such as TF-IDF, N-grams, and natural language processing (NLP) in the hope of improving detection accuracy. In this way, it also looks at how cyber threat intelligence and real-world datasets such as SpamAssassin, the UCI Machine Learning Repository, and PhishTank can be used to build strong models. The results show that ensemble learning and hybrid deep learning techniques are better at finding things than traditional methods, and they do this without increasing the number of false positives. Challenges such as adversarial SMS attacks, multilingual phishing messages, and real-time detection limitations remain plausible. Future works need to look into adaptability to real-time models, CTI-based threat analysis, and understandable AI (XAI) detection transparency. Applying machine learning-driven smishing detection brings up the overall solution's intelligent automated approach and adaptive defense mechanisms against mobile phone phishing threats evolving, resulting in increased security for mobiles and, hence, their users.

Keywords: Cyber Threat Intelligence, Machine Learning, Smishing Detection, SMS Spam Classification, Convolutional Neural Network, Long Short Term Memory, Term Frequency Inverse Document Frequency, Optimal Feature Extraction Algorithm, Independent Recurrent Neural Network, Capsule Network, Random Forest

Abbreviations:

SMS: Short Message Services
SVM: Support Vector Machines
CTI: Cyber Threat Intelligence
ML: Machine Learning
LSTM: Long Short-Term Memory

TD-IDF: Term Frequency Inverse Document Frequency
OFVA: Optimal Feature Extraction Algorithm
IndRNN: Independent Recurrent Neural Network
CapsNet: Capsule Network
RF: Random Forest
CTI-MURLD: Cyber Threat Intelligence-based Malicious URL Detection
LSTM: Long Short-Term Memory
BoW: Bag-of-Words
MLP: Multilayer Perceptron
CNN: Convolutional Neural Networks
AI: Artificial Intelligence
URL: Uniform Resource Locator
MTD: Mobile Threat Defense
QML: Qt Modeling Language
NLP: Natural Language Processing

I. INTRODUCTION

The onslaught of scamming is inescapable, what with us using our phones for everything these days. Smishing (SMS phishing) is one of the scariest threats. This involves attackers sending fraudulent messages appearing to come from trusted organizations like banks, delivery services, and government agencies. Such short messages either harass the receivers into clicking on malicious links or solicit sensitive information that opens a door to financial fraud, identity theft, or even malware infection. There is a need to detect such scams as they grow cleverer with time. Conventional smishing detection approaches such as blacklisting and rule-based filters are no longer effective since the attackers have learned to continuously change their arsenal. Machine learning (ML) algorithms are now being employed as robust devices to combat this major burgeoning issue of smishing message detection. ML models can learn from past evidence to seek patterns in smishing messages and identify suspicious messages, all the while enhancing themselves as a result of continuous learning.

Therefore, the main emphasis of this paper is on:

- How machine learning can help in the detection of smishing messages.
- A comparison of different ML models and their relative performance in detecting phishing.
- The issue of detecting new attacks and reducing false positives.
- Recommendations for the future development of AI-based SMS phishing detection with real-time data analysis.

The work is concerned with using "machine learning," a sub-parceling of AI incorporation, in the detection of smishing messages. This paper reviews various methods under machine learning for smishing detection in detail. The proposed methods cover neural networks, support vector machines (SVM), deep learning models, random

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*Correspondence Author(s)

Aqsa Shaikh, Student, Department of Information Technology, University of Mumbai, Mumbai (Maharashtra), India. Email ID: skaqsa242@gmail.com

Mariya Shaikh, Student, Department of Information Technology, University of Mumbai, Mumbai (Maharashtra), India. Email ID: shaikhmariya2909@gmail.com

Srivaramangai R.*, Head of the Department of Information Technology, University of Mumbai, Mumbai (Maharashtra), India. Email ID: rsrimangai@gmail.com, ORCID ID: [0000-0003-2723-6067](https://orcid.org/0000-0003-2723-6067)

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forest, and decision trees for the comparison of their efficiencies in distinguishing phishing messages. The paper discusses different methods of feature extraction, such as text-based methods (TF-IDF, N-grams), URL analysis, and a behavioural analysis that contribute to improving the accuracy of detection levels. We will talk about various machine-learning methods and how they can be useful for fighting against such kinds of text message schemes. In doing so, we aim to give a detailed overview of how this technology can be used against smishing and what challenges remain.

II. LITERATURE REVIEW

According to Timko et al. [1], the study evaluated the employability of users in differentiating between genuine messages and smishing messages through an online survey that included 187 participants. The implications of the findings suggest that attention and security behavioural scores significantly affect users' accuracy when identifying smishing messages. The author states that fake messages' accuracy was 67.1%, while real messages' accuracy was 43.6%, indicating the difficulty in the matter concerning user awareness. Mahmood and Hameed [2], there is an increasing threat of smishing, which involves phishing via SMS messages in the area of mobile communication. The study proposes a way to detect smishing that uses a combination of URL inspection by Google Engine and VirusTotal with an analysis of content in the SMS. The research used four machine-learning classifiers continuing from the classification of messages into legitimate vs. smishing using Support Vector Machine (SVM), Random Forest (RF), Adaptive Boosting (AdaBoost), and Extreme Gradient Boosting (XGBoost). The model thereby attained an accuracy of 98.5%, which is superior to existing methods concerning smishing message detection. Ankit et al. [3], Smishing—a growing threat to Cyber Security—deceives the users into installing malicious software. The approach is a two-phase machine learning methodology for spam and smishing text detection: first, spam is to be differentiated from ham, and afterward, smishing detection is to be performed on spam. Achieving 96% accuracy with different classifiers, notably neural networks. The study affirms the necessity of unique phishing features and information gain values for better-quality classification. Sharif et al. [4], detection of suspicious text is a very vital subject in cyber security, particularly in Bengali language processing. In his studies, he proposed a machine-learning-based model for classifying text into suspicious and non-suspicious. With a collection of 7000 Bengali text documents, the model, which encompasses classifiers like logistic regression and random forest, managed to give 84.57% accuracy. His study emphasizes that both unigram and bigram features are significant for enhancing classification performance. Mohanty et al. [5], identifying the cyber threats posed by suspicious URLs is vital for the internet security domain. It also creating unique models of multi-class ML-based classification to recognize the instance of spam, phishing, defacement attacks, and malware. A model using classifiers such as decision trees and logistic regression can attain an accuracy of 94.1% through lexical features of URLs. It is shown that feature extraction and classification techniques would significantly improve the

current state of threat detection. Sohn et al. [6], Currently, spam filtering has been taken a step up by stylistic features beyond plain content-based filtering. This study considers the usefulness of stylometric features like special-character usage, word length, part-of-speech n-grams, and function word frequency in spam and legitimate message discrimination. Using a machine-learning model, their method improves detection accuracy by avoiding false negatives while keeping a very low false-positive rate. Thus, the results indicate that the inclusion of writing style patterns has considerably improved the performance of SMS spam filters. Schlette et al. [7], Cyber Threat Intelligence (CTI) is important in a structured manner for managing security incident responses and their associated threat data in an organization. The study evaluates the six CTI formats identified as key concepts for increased efficiency in incident response. This involves discussing playbooks, automation, and standardization of the format in cybersecurity defense. The results hint at the fact that using several different CTI formats increases response capability while enabling organizations to mitigate cyber threats more effectively. Chong et al. [8], The malicious detection of URLs is necessary because increasing mobile device usage has led to the evolution of web vulnerabilities. Their efforts consider machine learning combining URL lexical features, JavaScript source code features, and payload size for malicious URL identification. Using an SVM with a polynomial kernel, they achieved 81% accuracy and a 74% F1 score. The work discusses the reassurance of real-time detection of malicious URLs, pointing out how feature extraction such as URL patterns and JavaScript obfuscation markers can hugely bolster security. Jain et al. [9], spam and phishing attacks are critical threats in cybersecurity. His work proposes machine learning-based spam messages and phishing link detection. The model is built depending on the datasets of known phishing sites and spam messages for newly incoming content, which is classified according to unique features. Various algorithms, including Random Forest and Support Vector Classifier, were evaluated and exhibited high capabilities in detecting threats. Jaiswal and Raut [10], Cybersecurity presents URL as a sizable threat due to scams, data theft, and spreading malicious programs. Hence, the study suggests machine learning to detect and block URLs. The model proposed improves detection by analyzing content through web crawling and sentiment analysis. The study portrays a pressing need for enhanced detection methodologies, as with the changes in the cyber world, the threats are evolving too. Aljabri et al. [11], Malicious URLs have posed an increasing threat in cybersecurity, resulting in phishing, malware, and data breaches. The review explores machine learning techniques for detecting these threats while analyzing the various types of features, detection methods, and limitations of datasets. The study argues that traditional blacklists cannot identify new threats and that supervised learning and deep learning models show good promise in detecting malicious URLs. The paper also discusses the challenges of dataset quality and the evolving nature of attacks, emphasizing the need for detection techniques that can learn and adapt. Tamal et al. [12], The growing nature of

phishing attacks poses this ever-increasing cybersecurity threat: their evolving nature and inertia of the existing detection techniques. This study aims to present the OFVA (Optimal Feature Extraction Algorithm) as a potential means of improving the existing phishing detection using supervised machine learning (ML). After 274,446 URLs were analyzed, 41 important intra-URL features were extracted from the model, and multiple classifiers were applied, among which random forests achieved an accuracy of 97.52%. The study argues for the need for adaptive detection methods in order to wage a war against phishing attacks efficiently. Rifah et al. [13], URL detection is crucial in safeguarding against a spectrum of cyber threats such as phishing and malware attacks. They propose a machine learning detection framework, which is based on logistic regression in classifying URLs as safe and unsafe. The system relies on analyzing URL structure and behavior for effective identification of links that may pose a threat without the use of webpage content. This method protects cyberspace via rapid and effective threat detection. Li and Dib [14], propose a new machine-learning architecture for detecting both known and unknown malicious URLs in real-time. The system classifies URLs into three major classes, namely phishing, malware, and others, leveraging tree-based algorithms and CL_K-means. It achieves 92.54% accuracy regarding zero-day attacks, completing a single classification in less than 14 milliseconds. Yuan et al. [15], The model of a parallel neural joint is proposed for malicious URL detection. The above system converts URLs into word embeddings and grayscale images, then extracts semantic and visual features and processes them with IndRNN and CapsNet with an attention mechanism. This achieves a very high classification accuracy and outperforms traditional techniques. Xuan et al. [16], It proposes a machine learning-based method to detect malignant URLs. The system extracts feature from lexical features, host-based features, and correlated groups to create new URL attributes to train Random Forest, SVM, etc., to categorize it as safe or malignant. The outcome proved that the Random Forest Model can achieve high accuracy on larger datasets, which makes this approach not only efficient but also practical. Ravindra et al. [17], phishing creates another big security threat in cybersecurity by fooling the users to get sensitive information from them. The study shows the Random Forest machine learning-based detection system that will classify a URL either as legitimate or phishing. The system analyzes URL features such as length, suspicious characters, and subdomains to improve detection accuracy. With the dataset of 4000 URLs, this model has achieved an accuracy of 86% to provide effective identification and blocking of phishing threats. Cao and Caverlee [18], Social media spam URL detection holds importance in mitigating phishing and malware attacks. The research describes a type of detection based on behavior that studies the posting patterns of URLs along with user interactions with those URLs. By measuring and evaluating fifteen behavioral features using a dataset of seven million URLs, the approach evaluates with the highest accuracy at 86 percent precision and recall. This method strengthens spam detection beyond the traditional techniques that rely on blacklists. Reyes-Dorta et al. [19], in their study, evaluated ML and QML techniques for detecting fraudulent URLs.

More than 90% true positive rates were achieved using classical ML algorithms like decision trees, logistic regression, and neural networks. Subsequently, QML was explored through methods such as the Variational Quantum Classifier, yielding promising results that seem to match those achieved by classical models. The authors highlight the promise of QML in cybersecurity, albeit limited due to the constraints of current quantum hardware and datasets. Sonowal [20], improving smishing message (SMS phishing detection) can happen by feature selection and machine learning. Conducted five different ranking algorithms and concluded that the one with the best performance is Kendall rank correlation with an AdaBoost classifier, which showed the highest accuracy of 98.40% by reducing the feature set to 61.53%. Thus, indicating an efficient way of smishing detection improvement. Canali et al. [21], In essence, the goal of Prophiler is to quickly filter out malicious web pages in the most efficient manner possible. This system works with static analysis techniques to identify malicious features found in the HTML content, JavaScript code, and URL structure of a particular site. The classification mechanism separates benign from malicious web pages by machine learning classifiers. Prophiler significantly reduces the workload of dynamic analysis tools by filtering about 85 percent of the benign pages. In the study, they show how Prophiler achieves high accuracy in detecting actual threats while yielding a low false negative rate. This makes it truly usable for large-scale web security. Tabassum et al. [22], URLs that are malevolent pose threats to cybersecurity because of phishing websites and malware. The traditional blacklisting methods have proved to be ineffectively slow at best as far as new threats are concerned. It is against this backdrop that the study investigates machine learning for better accuracy in the detection of harmful URLs. Its findings point to Random Forest and Neural Networks achieving above 90% accuracy in various tests. With regard to what the research had to offer, it also highlighted things such as the evolving nature of threats and challenges faced because of zero-day attacks and proposed adaptations in improvement for future results. Ghaleb et al. [23], Malicious websites are a big cybersecurity device, where they are often used for phishing, malware, and fraud. It's really hard to detect new kinds of attacks using the old-style detection approaches. In this paper, we present Cyber Threat Intelligence-based Malicious URL Detection (CTI-MURLD), an ensemble learning-based model to improve detection accuracy. It extracts URL-based cyber threat intelligence (CTI) and Whois-based features based on a web search. The best classification improves detection accuracy through two-stage classification between Random Forest (RF) and Multilayer Perceptron (MLP). Improvement in accuracy is 7.8%, with a decrease in false positives by 6.7% compared with previous methods. Das Gupta et al. [24], An increase in the instances of SMS being used as an avenue for fraud or sending unsolicited messages has made this study relevant. Conventional rule-based methods of detection would often fail in dealing with new spam patterns. This study proposes the use of machine learning to classify SMS as either spam or ham (legitimate). Several classifiers, including Random Forest (RF), Naïve

Bayes, and Support Vector Machine (SVM), were tested to compare the accuracy of spam detection. Results showed that Random Forest had the best accuracy in classifying spam messages. The study points out that the success of spam detection is influenced significantly by feature selection and text preprocessing. Palwankar et al. [25], detected the malicious link to mitigate phishing and malware attacks [26]. Traditional methods have become a hiccup as they were unsuccessful to updating with the threats occurring [27]. This research suggests an approach based on machine learning, where logistic regression and random forest models are used [28]. The system will be URL-feature-based and not rely on the content of the page. Improvements in operational efficiency will also look at threats by integrating VirusTotal and GNews APIs for real-time verification and updating [29]. The results produce improvements in both accuracy and security that will make browsing safer [30].

III. OBSERVATIONS

A. Increasing Smishing Attacks Threatening Growth

Smishing/SMS Phishing—Having serious cyber-attack potential, smishing also employs social engineering to coerce users into divulging sensitive information, downloading malware, or visiting bogus sites. In the modern world, where mobile communication is increasingly becoming an important part of daily life, such cybercriminals rely more on trust in text messaging services instead of going for individuals. Attackers make legitimate communications look like they are from banks, government, delivery services, or well-known brands to establish their trust with users. Mobile banking and online transactions have only accelerated the ease at which financial scams, identity thefts, or ransomware attacks can perpetrate new victims. Some of the mainstream rule-based detection methods—like blacklists and keyword-based filtering—do not manually capture the latest evolving attack strategies. This fact renders them practically ineffective against so-called sophisticated phishing schemes.

B. Machine Learning Role in the Efficiency of Smishing Detection

Machine Learning (ML) is a potent and well-streamlined mechanism in counter-puncturing smishing cases since one can employ automated and intelligent detection schemes adaptable to new threats. ML models, based on analyzing message content, URLs, the behavior of the sender, and metadata, improve security through their in-depth scrutiny of troubling suspicious messages. It is a reality that supervised learning models, like support vector machines, random forests, decision trees, and neural networks, have been able to offer highly accurate results in smishing detection. Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) are deep-learning models having excellent success rates in understanding language patterns and providing contexts of messages, therefore positively affecting the classification of phishing messages. Combined with ensemble learning and deep learning, considered as Hybrid approaches integrate the different techniques this superiority over unimodal approaches exhibits by dropping false positives and raising detection rates. AI-backed real-time dynamic detection systems combined with

reinforcement learning are set to ensure dynamic protection by enriching knowledge bases with updated phishing patterns.

C. Feature Engineering and Data Analysis Significance

Extracting and analyzing relevant features extracted from text messages, URLs, and sender information is a fundamental aspect of improving detection rates for smishing.

- *Lexical Feature Extraction:* TF-IDF, N-grams, or Bag-of-Words (BoW) would perform extremely effective feature extraction, helping ML models analyze the structure of an SMS in order to detect abnormalities associated with phishing messages.
- *URL Inspection:* Analysis would reveal details from IP address, domain age, and WHOIS information with the use of URL-based features that can also help classify the fraudulent links-abusive links embedded in SMS messages.
- *Behavioral Analysis:* Tracking message patterns, response frequencies from senders, and behavior will allow ML models to distinguish whether it's potential phishing attempts or just simple communication.
- *Threat intelligence:* CTI—Integrating with the external database on the cybersecurity threat from VirusTotal, PhishTank, and Google Safe Browsing will further improve how detection systems identify malevolent URLs or phishing attempts.

D. Challenges in Smishing Detection

Nonetheless, some challenges remain, even with the successful machine learning-based deployment for detecting smishing.

- *Adversarial SMS Attacks:* Constant metamorphosis is done in their SMS by the attackers to avoid detection from the ML. They use different obfuscation techniques such as replacing characters, adding spaces, and varying words.
- *Multilingual Contextual Issues:* Many phishing messages come in several languages, which means it becomes tough for models to generalize well if they are trained on specific linguistic features.
- *Real-Time Processing Constraints:* Detect the smishing messages and block them in real time while achieving a very good detection accuracy with the lowest rates of false positives for security systems on mobile devices.
- *Privacy and Data Security Problems:* For training machine learning models through SMS data, one needs to access user messages, through which privacy issues and compliance with several data protection regulations like GDPR come into the scene.

E. Future Research and Emerging Trends

Future research will be directed to further developing security measures that are increasingly smarter, adaptable, and dynamic.

- *Explainable AI:* Making decision-making in an ML model interpretable will help in flagging a

message as malicious so that security teams and users can understand why something is flagged as malevolent.

- *Federated Learning for Detection Preservation of Privacy*: SMS data won't be popping up in a centralized database for training; models will train locally on the user device to make it safer than a whole centralized system.
- *Adaptive and Self-Learning Models*: Reinforcement learning techniques will support continuous learning and adaptation to new phishing attack patterns.
- *Mobile Threat Defense (MTD)*: Smishing detection might then be combined with larger cyber strategies, including endpoint protection, fraud detection, and behavior biometrics, to give round-the-clock security solutions.
- *Automated Incident Response*: Mobile security application integration will speed up and automate the remediation of smishing.

IV. CONCLUSION

Smishing attacks are serious risks to cybersecurity, requiring countermeasures far advanced beyond traditional filtering techniques. Machine-learning-based approaches have shown high degrees of accuracy in the detection of smishing by use of deep learning, behavioral analysis, and real-time threat intelligence. Machine learning models, namely supervised, unsupervised, and deep learning-based techniques, have successfully found application in the detection and mitigation of smishing attacks. Many methodologies for feature extraction—NLP, URL analysis, and behavioral monitoring—improved specificity for false positives. Nevertheless, the issues posed when faced with adversarial evasion attacks, multilingualism, and real-time processing availability constrain continuous model improvement and research over time. The blend of explainable AI, cyber threat intelligence, and models with adaptive machine learning constitutes the keystone for designing a robust, scalable, and transparent detection mechanism for smishing. By enhancing the research agenda on innovative threat detection, the cybersecurity fraternity can truly shove the needle in curbing the risk SMS phishing poses and towards improving mobile security overall.

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AUTHOR'S PROFILE



Aqsa Shaikh is a student of PG program M.S(Cyber Security) from the University Department of Information Technology, University of Mumbai. Aqsa has been proactively involved in solving Hackathon problems. She is proficient in programming languages such as Java, Python, jQuery and have worked in development of portal with MYSQL as the backend. Her flair for Cyber Security with initial skills on tools like Wireshark, NMAP etc., led her to pursue a masters in Cyber Security more focusing on various types of attacks and detection. Her expertise includes Cryptography and Network Security, Information Security. Currently she is focusing on "Smishing Detection" as her research project area.



Mariya Shaikh is a student of PG program M.S(Cyber Security) from the University Department of Information Technology, University of Mumbai. Mariya has a good exposure to the current trends in IT industry mainly the latest developments in the field of Cyber Security. She has a varied skillset ranging from MS SQL Server, PHP, JSP for web development and Python for research. She has actively participated in Hackathons and Project Exhibitions conducted by various educational institutions. She is currently focused with her research project on Smishing Detection and is planning to appear for Cyber Security certifications. Her expertise includes Cryptography and Network Security, Information Security.



Srivaramangai R., Head, Department of IT, University of Mumbai, India. Having 24 years of experience in teaching and 6 years in industry. The specialization areas are artificial intelligence, security, image processing. Has industry experience in web development and report code generators. Has published more than 35 International journal papers, 25 conference papers, resource persons for various workshops and chaired sessions. The papers relevant to Cyber Security includes "Assessment of Deep Packet Inspection System of Network traffic and Anomaly Detection", "Enhancing Security using ECC in Cloud Storage", "Recapitulation of the Use of Machine Learning for Prevention of DDoS Attack on SDN Controller" and "Unmasking Deceptive Websites : Harnessing Machine Learning For Phishing Detection".

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